WORKED SOLUTION

STEP II 2004, Question 7

Pure • Interval Bisection

STEP II 2004, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q7

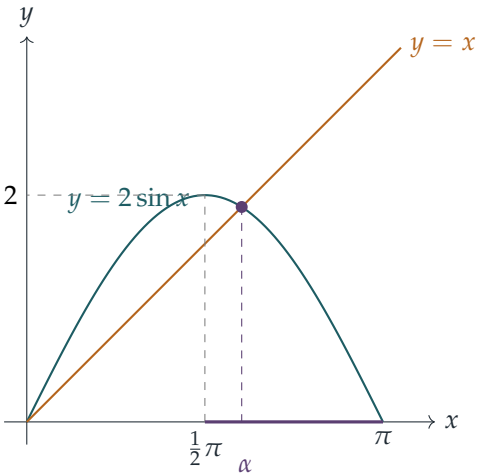
The key idea

A root of $2\sin x - x = 0$ is a place where the curve $y = 2\sin x$ meets the line $y = x$, so the graphical part is about one sine curve and one straight line. The bisection steps then come down to one question each time: is $2\sin c$ bigger or smaller than c at the midpoint? The half-angle formula turns the sines of $\frac{5}{8}\pi$ and $\frac{9}{16}\pi$ into nested square roots, and comparing *squares* of two positive numbers keeps the arithmetic to the given approximations.

Solution

Exactly one root in I_0

We have $f(x) = 0 \iff 2\sin x = x$, so the roots are the x-coordinates of the points where $y = 2\sin x$ meets $y = x$.



On $[\frac{1}{2}\pi, \pi]$:

- at $x = \frac{1}{2}\pi$ the curve is at height 2 and the line at height $\frac{1}{2}\pi < 2$ (since $\pi < 4$), so the curve is above the line;
- at $x = \pi$ the curve is at height 0 and the line at height $\pi > 0$, so the curve is below the line;
- the curve $y = 2\sin x$ is decreasing on this interval while the line $y = x$ is increasing.

Both graphs are continuous, so the curve must cross the line somewhere in the interval; and since one is falling and the other rising, once the curve is below the line it can never come back above it. So they cross exactly once, and $f(x) = 0$ has exactly one root α in I_0. In terms of f: $f(\frac{1}{2}\pi) = 2 - \frac{1}{2}\pi > 0$, $f(\pi) = -\pi < 0$, and f is strictly decreasing there.

In every bisection step below, we keep the half on which f changes sign. Since $f(\frac{1}{2}\pi) > 0$, the sign of $f(c_n)$ decides everything: if $f(c_n) < 0$ the root is to the left of c_n, and if $f(c_n) > 0$ it is to the right.

Finding I_1

Here $c_0 = \frac{3}{4}\pi$ and

$$f(\tfrac{3}{4}\pi) = 2 \cdot \frac{1}{\sqrt{2}} - \tfrac{3}{4}\pi \approx 1.4 - \tfrac{3}{4}\sqrt{10}.$$

Since $\sqrt{10} > 3$, we have $\frac{3}{4}\sqrt{10} > 2.25 > 1.4$, so $f(c_0) < 0$. With $f(a_0) > 0$ this gives $f(a_0)f(c_0) < 0$, so

$$I_1 = [\tfrac{1}{2}\pi, \tfrac{3}{4}\pi].$$

Finding I_2

Now $c_1 = \frac{1}{2}(\frac{1}{2}\pi + \frac{3}{4}\pi) = \frac{5}{8}\pi$, and we need the sign of $2\sin\frac{5}{8}\pi - \frac{5}{8}\pi$.

> **?** **Ask yourself** How can we find $\sin\frac{5}{8}\pi$ without a calculator?

Use $\sin\frac{5}{8}\pi = \sin(\pi - \frac{5}{8}\pi) = \sin\frac{3}{8}\pi = \cos\frac{1}{8}\pi$, and then the double-angle formula $\cos 2\theta = 2\cos^2\theta - 1$ with $\theta = \frac{1}{8}\pi$:

$$4\cos^2\tfrac{1}{8}\pi = 2\big(1 + \cos\tfrac{1}{4}\pi\big) = 2 + \sqrt{2}.$$

So $(2\sin\frac{5}{8}\pi)^2 = 2 + \sqrt{2}$. Both $2\sin\frac{5}{8}\pi$ and $\frac{5}{8}\pi$ are positive, so we may compare their squares:

$$(2\sin\tfrac{5}{8}\pi)^2 = 2 + \sqrt{2} \approx 3.4, \qquad (\tfrac{5}{8}\pi)^2 \approx \tfrac{25}{64} \cdot 10 = \tfrac{250}{64} \approx 3.9.$$

Hence $2\sin\frac{5}{8}\pi < \frac{5}{8}\pi$, that is, $f(c_1) < 0$. Again $f(a_1) = f(\frac{1}{2}\pi) > 0$, so $f(a_1)f(c_1) < 0$ and

$$I_2 = [\tfrac{1}{2}\pi, \tfrac{5}{8}\pi],$$

as required.

Finding I_3

Now $c_2 = \frac{1}{2}(\frac{1}{2}\pi + \frac{5}{8}\pi) = \frac{9}{16}\pi$. As before, $\sin\frac{9}{16}\pi = \sin\frac{7}{16}\pi = \cos\frac{1}{16}\pi$, and the double-angle formula with $\theta = \frac{1}{16}\pi$ gives

$$(2\sin\tfrac{9}{16}\pi)^2 = 4\cos^2\tfrac{1}{16}\pi = 2\big(1+\cos\tfrac{1}{8}\pi\big) = 2+2\cos\tfrac{1}{8}\pi = 2+\sqrt{2+\sqrt{2}},$$

using $2\cos\frac{1}{8}\pi = \sqrt{2+\sqrt{2}}$ from the previous step. With $\sqrt{2} \approx 1.4$, we have $\sqrt{2+\sqrt{2}} \approx \sqrt{3.4} > 1.8$ (because $1.8^2 = 3.24$), so

$$(2\sin\tfrac{9}{16}\pi)^2 > 3.8, \qquad (\tfrac{9}{16}\pi)^2 \approx \tfrac{81}{256}\cdot 10 = \tfrac{810}{256} < 3.2.$$

So $2\sin\frac{9}{16}\pi > \frac{9}{16}\pi$, that is, $\mathrm{f}(c_2) > 0$. This time $\mathrm{f}(a_2)\mathrm{f}(c_2) > 0$, while $\mathrm{f}(c_2)\mathrm{f}(b_2) = \mathrm{f}(\frac{9}{16}\pi)\,\mathrm{f}(\frac{5}{8}\pi) < 0$. Therefore

$$I_3 = [\tfrac{9}{16}\pi, \tfrac{5}{8}\pi].$$

The approximations are safe here: at each midpoint the two quantities being compared differ by about 0.5 or more, far more than the error introduced by $\frac{1}{\sqrt{2}} \approx 0.7$ or $\pi \approx \sqrt{10}$.

Checks

- Running the bisection numerically gives the same intervals, and the root is $\alpha \approx 1.8955$, which lies between $\frac{9}{16}\pi \approx 1.767$ and $\frac{5}{8}\pi \approx 1.963$.
- Exact values agree with the approximate ones: $\mathrm{f}(\frac{3}{4}\pi) \approx -0.942$, $\mathrm{f}(\frac{5}{8}\pi) \approx -0.116$ and $\mathrm{f}(\frac{9}{16}\pi) \approx 0.194$.
- Each interval has half the length of the one before: $\frac{1}{2}\pi$, $\frac{1}{4}\pi$, $\frac{1}{8}\pi$, $\frac{1}{16}\pi$.

Common slips

- Showing a sign change but not explaining why there is only *one* root. A sign change alone gives at least one; you need the decreasing-versus-increasing argument for uniqueness.
- Comparing $2\sin c$ and c by squaring without first noting that both are positive.

Beyond STEP

The sign-change argument is the intermediate value theorem, one of the first big theorems in a university analysis course; its proof needs the completeness of the real numbers, and bisection is essentially that proof run as an algorithm. Bisection always converges, but slowly (one binary digit per step). Methods such as Newton–Raphson converge much faster, but can fail, so numerical analysis is largely about trading safety against speed. For the theory, see Abbott, *Understanding Analysis*.

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